

# AKSHIT PRATAP SINGH



## Computer Science & Engineering (AI & ML) Undergraduate | Cybersecurity | AI/ML | Software Development

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## PROFESSIONAL SUMMARY

Computer Science and Engineering undergraduate specializing in Artificial Intelligence and Machine Learning, with hands-on learning exposure in cybersecurity, network traffic analysis, AI-powered threat detection, software development, hackathons, and student outreach. Interested in building practical, documented solutions across cybersecurity, AI/ML, and software engineering.

## EDUCATION

### Bachelor of Technology (B.Tech) — Computer Science & Engineering, Specialization in Artificial Intelligence and Machine Learning

United Group of Institutions | Dr. A.P.J. Abdul Kalam Technical University (AKTU)

### Senior Secondary Education — Physics, Chemistry, Mathematics and Physical Education

CBSE

## EXPERIENCE & LEADERSHIP

### Cybersecurity Trainee & Project Intern

United Group of Institutions

- Worked on network traffic analysis and threat-identification learning activities.
- Used Wireshark for packet capture, filtering, protocol inspection, and network-flow analysis.
- Reviewed suspicious traffic patterns and documented observations in a controlled learning environment.

### Internshala Student Partner (ISP 59)

Internshala

- Support student awareness of internships, training programs, and career-development opportunities.
- Contribute to outreach, communication, networking, and student engagement activities.

### Campus Ambassador — Tata Imagination Challenge

Tata Group

- Promote awareness of the challenge through student-facing communication and outreach.
- Support participation awareness, networking, and campaign-related engagement.

### College Ambassador — Techfest, IIT Bombay

Techfest, Indian Institute of Technology Bombay

- Support event awareness and outreach among students.
- Communicate information about technology-focused events and participation opportunities.

### Campus Ambassador — Air India Group

Air India Group

- Support student outreach, communication, and awareness activities.
- Represent program information through student-focused engagement.

## SELECTED PROJECTS

### NetSentinel AI — AI-Powered Network Threat Detection System

IGNITION Innovation Hackathon | Team AegisStrik | Separate project repository

- Developed an AI-powered network threat-detection concept for Security Operations Center workflows.
- Combined a Random Forest machine-learning approach with rule-based detection for suspicious network activity.
- Contributed to technical coordination, project planning, risk-oriented security insights, and presentation preparation.

### Network Traffic Analysis Framework for Cyber Attack Detection

Cybersecurity training project | Separate project repository

- Built and documented a controlled security-testing environment using Kali Linux and Windows.
- Used Wireshark for packet inspection and reviewed network telemetry during authorized lab exercises.
- Worked with Nmap, Hping3, and Hydra only in controlled, authorized learning scenarios.
- Documented observations, limitations, and defensive security considerations.

### SmartPose AI Camera

Android / Computer Vision project

- Explored a camera application concept using CameraX-oriented workflows and pose-guidance functionality.
- Organized the proposed application structure around MVVM and Clean Architecture concepts.
- Documented setup, permissions, extension points, and customization guidance.

**Flying Cat**

*Interactive game prototype*

- Developed and documented an interactive game prototype with gameplay visuals, controls, and a desert-themed environment.
- Prepared project architecture notes, setup guidance, controls, and customization documentation.

**College Smart Issue Reporting System**

*U-HACK 4.0 | Team CODEX CREATORS*

- Contributed to a campus issue-reporting concept in a time-constrained hackathon environment.
- Worked on rapid prototyping, teamwork, system planning, and technical presentation.

**TECHNICAL SKILLS**

|                   |                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Programming       | Python, C, C++, JavaScript, HTML, CSS, Java                                                                                   |
| Cybersecurity     | Network Security, Network Traffic Analysis, Packet Inspection, Threat Analysis, Authorized Security Testing                   |
| Security Tools    | Wireshark, Nmap, Hping3, Hydra, Kali Linux, Windows, Linux                                                                    |
| AI/ML & Data      | Artificial Intelligence, Machine Learning, Random Forest, Computer Vision, Pose Estimation, Generative AI, Prompt Engineering |
| Development Tools | Git, GitHub, React, Vite, Tailwind CSS, Android Studio, CameraX concepts                                                      |
| Engineering       | MVVM, Clean Architecture, Modular Design, Separation of Concerns, Rapid Prototyping, Technical Documentation                  |
| Professional      | Problem Solving, Team Collaboration, Leadership, Project Coordination, Technical Presentation, Public Speaking, Networking    |

**RELEVANT CERTIFICATIONS & LEARNING CREDENTIALS**

| Credential                                                               | Details / Relevance                                                                                                                          |
|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| CS50x: Introduction to Computer Science                                  | Harvard University — foundational computer science, programming, algorithms, and problem-solving learning.                                   |
| Summer Training in Network Traffic Analysis & Cyber Attack Detection     | United Group of Institutions — network traffic analysis, Wireshark, packet inspection, protocol analysis, and controlled security exercises. |
| 2-Hour Ethical Hacking & Cybersecurity Workshop                          | VaultofCodes — introductory ethical hacking and cybersecurity concepts.                                                                      |
| Executive Certification in Artificial Intelligence                       | UniAthena — artificial intelligence concepts and executive-level AI learning.                                                                |
| Cambridge International Qualification: Basics of Artificial Intelligence | Foundational AI concepts and applications.                                                                                                   |
| Introduction to Generative AI Studio                                     | Simplilearn — introductory generative AI learning.                                                                                           |
| Introduction to Prompt Engineering                                       | Simplilearn — prompt design and interaction with generative AI systems.                                                                      |
| 24-Hour Innovation Hackathon                                             | U-HACK 4.0: 24-Hour Innovation Hackathon                                                                                                     |
| IGNITION Innovation Hackathon                                            | IGNITION Innovation Hackathon organized by Concept2Action (C2A)                                                                              |
| Use Canva to Design Digital Course Collateral                            | Canva — digital design and course-collateral creation.                                                                                       |
| Google Ads Certifications                                                | Google Ads — introductory digital advertising and campaign concepts.                                                                         |
| HP LIFE: AI for Business                                                 | HP LIFE — introductory business applications of artificial intelligence.                                                                     |
| HP LIFE: 3D Printing                                                     | HP LIFE — introductory 3D-printing concepts.                                                                                                 |

**ACHIEVEMENTS & PROFILES**

- Codolio profile: @OxAPEX\_DEVL :100+ questions solved and 15+ active days.
- Coding practice across LeetCode, HackerRank, CodeChef, GeeksforGeeks, Codeforces, and GitHub.
- Participation in hackathon-based work through Team AegisStrik and Team CODEX CREATORS.

**LANGUAGES**

- English — Professional working proficiency.
- Hindi — Native proficiency.
- French — Basic proficiency.

## INTERESTS

Cybersecurity research, AI/ML model building, open-source development, technical content creation, hackathons, and emerging technologies.